

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester V Examination, 2022-23

Paper No. MTB-511: Probability

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1.

- (a) Explain the axiomatic definition of probability. [2]
- (b) Define σ -algebra with a suitable example. [2]
- (c) If ϕ is the empty set, prove that $P(\phi) = 0$. [2]
- (d) If X is a binomial random variable with $E(X) = 2$ and $\text{Var}(X) = 4$, evaluate $P(X = 5)$. [2]
- (e) Define the cumulative distribution function. [2]
- (f) A card is drawn at random from a well-shuffled deck. Find the probability that it is a Spade or a King. [2]
- (g) Define the partition of a sample space with an example. [2]
- (h) Define entropy $H(X)$ and sample mean $M_n(X)$. [2]
- (i) Explain mutual information $I(X; Y)$ for bivariate random variables. [2]
- (j) Define joint density function and marginal density function for bivariate random variables. [2]
- (k) If $X \sim N(\mu, \sigma^2)$, prove that all odd-order central moments are zero, i.e., $\mu_{2r+1} = E[(X - \mu)^{2r+1}] = 0$ for $r = 0, 1, 2, \dots$ [2]

Question 2.

- (a) Let a random variable X have the following probability distribution:

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Find (i) the value of k , (ii) $P(1.5 < X < 4.5 \mid X > 2)$, and (iii) the CDF of X . [4]

- (b) The CDF of a random variable X is given by $F(x) = 0$ if $x < 0$; $F(x) = 1 - \frac{(3-x)^2}{9}$ if $0 \leq x < 3$; $F(x) = 1$ if $x \geq 3$. Find the PDF of X and evaluate $P(|X| < 1)$ using both PDF and CDF. [4]
- (c) The joint density function of X and Y is $f(x, y) = 6(1-x-y)$ for $x \geq 0, y \geq 0, x+y < 1$, and 0 otherwise. Find the marginal density functions and verify whether X and Y are independent. [4]
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Question 3.

- (a) A first box contains a white and b black balls; a second box contains c white and d black balls. One ball is transferred from the first to the second, then one ball is drawn from the second. Find the probability that it is white. [2]
- (b) The PDF of a random variable X is $f(x) = kx(2-x)$ for $0 < x < 2$. Find k , the mean, variance, and CDF. [4]
- (c) State and prove the Central Limit Theorem. [6]
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Question 4.

- (a) Prove that the Poisson distribution is a limiting case of the binomial distribution. [4]
- (b) Find the MGF of a random variable X uniformly distributed on (a, b) and hence find its mean and variance. [4]
- (c) If $X \sim \text{Exp}(\lambda)$, find its mean, variance, and MGF. [4]
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Question 5.

- (a) Let X, Y, Z be jointly continuous with joint PDF $f(x, y, z) = k(x + 2y + 3z)$ for $0 < x, y, z < 1$, and 0 otherwise. Find k and the marginal PDFs of X, Y, Z . [4]
- (b) If X is binomial(n, p), prove that its mean is $\mu = np$, variance $\sigma^2 = npq$, and that the central moments satisfy the recurrence $\mu_{r+1} = pq \left[r\mu_{r-1} + \frac{d\mu_r}{dp} \right]$. [4]
- (c) Prove the memoryless property of the exponential distribution. [4]
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Question 6.

- (a) The joint density of X and Y is $f(x, y) = Cxy$ for $0 < x < 1, 0 < y < 1$, and 0 elsewhere. Find: (i) C ; (ii) marginal densities of X and Y ; (iii) whether X and Y are independent; (iv) $P(X + Y \leq 1)$. [6]
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- (b) If $X \sim N(\mu, \sigma^2)$, prove that $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$. Also determine the MGF $M_X(t)$. [6]
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Question 7.

- (a) Let the joint PDF of (X, Y) be $f(x, y) = \frac{8}{15}(x + y^2)$ for $0 < x < 1$, $0 < y < 1$, and 0 otherwise. Obtain the marginal PDF of X , the conditional PDF of Y given $X = 0.8$, and evaluate $E(Y | X = 0.8)$. [6]
- (b) If X has PDF $f(x) = \frac{1}{2}e^{-|x|}$ for $-\infty < x < \infty$, show that the MGF of X is $M_X(t) = \frac{1}{1 - t^2}$, which exists for $|t| < 1$. Hence find the mean and variance of X . [6]